MCE5919: Gaussian Process for Machine Learning and Signal Processing

Lecture 7

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Overview

1. Motivation and the Kernel Trick

2. Reproducing Kernel Hilbert Spaces

3. Representer Theorem and KRR

4. Connection to Gaussian Processes

5. Summary

Why Nonlinear Models?

- Most real regression/classification problems are nonlinear
- Linear models underfit; high-degree polynomials suffer from the curse of dimensionality.
- Cover's Theorem (1965): mapping inputs to a sufficiently high-dimensional space makes the problem *linearly* separable with high probability.

Challenge: In a K-dimensional space, computations cost $O(K)$ — infeasible when $K \to \infty$.

Today's route: Kernel trick & RKHS $\to$ Representer theorem & KRR $\to$ GP connection

From Basis Expansion to the Kernel Trick

A classical strategy for nonlinear modelling:

1. Map $\boldsymbol{x} \in \mathbb{R}^I$ to a feature vector $\phi(\boldsymbol{x}) \in \mathbb{R}^K$ ($K \gg I$, possibly $K = \infty$).
2. Fit a linear model in the new space: $f(\boldsymbol{x}) = \boldsymbol{\theta}^T \phi(\boldsymbol{x})$.

Problem: explicit computation of $\phi(\boldsymbol{x})$ is expensive when K is large.

Key observation: most learning algorithms only need *inner products* $\langle \phi(\boldsymbol{x}_i), \phi(\boldsymbol{x}_j) \rangle$.
Define the kernel function:

$$\kappa(\boldsymbol{x}, \boldsymbol{x}') := \langle \phi(\boldsymbol{x}), \phi(\boldsymbol{x}') \rangle.$$

Evaluate κ in the original low-dimensional space — no need to form ϕ explicitly.

The 4-Step Kernel Workflow

1. Map implicitly: $\boldsymbol{x}_n \mapsto \phi(\boldsymbol{x}_n) \in \mathbb{H}$ for each training point.
2. Solve linear: formulate estimation as a linear problem in $\mathbb{H}$.
3. Re-express: write all computations in terms of inner products $\langle \phi(\boldsymbol{x}_i), \phi(\boldsymbol{x}_j) \rangle$.
4. Kernelise: replace every inner product by $\kappa(\boldsymbol{x}_i, \boldsymbol{x}_j)$.

Result: the nonlinear predictor costs only $O(N^2)$ kernel evaluations — independent of the (possibly infinite) dimension of $\mathbb{H}$.

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RKHS: Formal Definition

Let $\mathbb{H}$ be a Hilbert space of real-valued functions on $\mathcal{X} \subseteq \mathbb{R}^I$.

Definition (RKHS)

$\mathbb{H}$ is a Reproducing Kernel Hilbert Space if there exists $\kappa : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ such that:

1. For each $\boldsymbol{x} \in \mathcal{X}$: $\kappa(\cdot, \boldsymbol{x}) \in \mathbb{H}$ (*kernel atoms live in the space*).
2. Reproducing property: for all $f \in \mathbb{H}$, $\boldsymbol{x} \in \mathcal{X}$:

$$f(\boldsymbol{x}) = \langle f, \kappa(\cdot, \boldsymbol{x}) \rangle_{\mathbb{H}}.$$

Intuition: evaluating f at $\boldsymbol{x}$ is just an inner product with the kernel "atom" centred at $\boldsymbol{x}$.

Properties of RKHS

- Feature map: $\phi : \boldsymbol{x} \mapsto \kappa(\cdot, \boldsymbol{x}) \in \mathbb{H}$. Then:

$$\langle \phi(\boldsymbol{x}), \phi(\boldsymbol{x}') \rangle_{\mathbb{H}} = \kappa(\boldsymbol{x}, \boldsymbol{x}').$$

- Positive definiteness (Mercer / Aronszajn): κ is a valid kernel $\Leftrightarrow$ the kernel matrix K with $K_{ij} = \kappa(\boldsymbol{x}_i, \boldsymbol{x}_j)$ is positive semi-definite for any finite collection of points.
- Span:

$$\mathbb{H} = \overline{\text{span}}\{\kappa(\cdot, \boldsymbol{x}) \mid \boldsymbol{x} \in \mathcal{X}\}.$$

  The RKHS is completely generated by κ; given κ and $\mathcal{X}$, $\mathbb{H}$ is fully determined.

Common Kernel Functions

Name	Formula	RKHS dim.
Gaussian (RBF)	$\exp\left(-\frac{\|\boldsymbol{x}-\boldsymbol{x}'\|^2}{2\sigma^2}\right)$	∞
Polynomial	$(\boldsymbol{x}^T\boldsymbol{x}' + c)^r,\ c \geq 0$	finite
Laplacian	$\exp(-t\|\boldsymbol{x} - \boldsymbol{x}'\|),\ t > 0$	∞
Sinc (bandlimited)	$\mathrm{sinc}(x - x')$	∞

Construction rules: sums, products, and positive scalings of valid kernels are still valid kernels.

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Regularized Learning in RKHS

Given data $\{(\boldsymbol{x}_n, y_n)\}_{n=1}^N$, consider:

$$\min_{f \in \mathbb{H}} \; J(f) = \underbrace{\sum_{n=1}^{N} \mathcal{L}(y_n,\, f(\boldsymbol{x}_n))}_{\text{data fit}} + \lambda \; \underbrace{\Omega\big(\|f\|_{\mathbb{H}}^2\big)}_{\text{regulariser}},$$

where $\Omega : [0, +\infty) \to \mathbb{R}$ is strictly increasing, $\mathcal{L}$ is any loss function, and $\lambda > 0$.

Key challenge: minimising over all $f \in \mathbb{H}$ seems intractable when $\mathbb{H}$ is infinite-dimensional.

Representer Theorem

Theorem

The minimiser of $J(f)$ over $\mathbb{H}$ admits the finite expansion:

$$f^*(\cdot) = \sum_{n=1}^{N} \theta_n \, \kappa(\cdot, \boldsymbol{x}_n), \quad \theta_n \in \mathbb{R}.$$

Proof sketch: Decompose $f = f_A + f_\perp$ where $f_A \in A := \operatorname{span}\{\kappa(\cdot, \boldsymbol{x}_n)\}$ and $f_\perp \perp A$.

- Reproducing property: $f(\boldsymbol{x}_n)$ depends only on f_A (not $f_\perp$).
- But $\Omega(\|f\|^2) = \Omega(\|f_A\|^2 + \|f_\perp\|^2) \geq \Omega(\|f_A\|^2)$.
- Any $f_\perp \neq 0$ only *increases* cost $\Rightarrow f_\perp^* = 0$. $\square$

Takeaway: optimization over $\mathbb{H}$ reduces to $\boldsymbol{\theta} \in \mathbb{R}^N$.

Representer Theorem: Significance

Remarks

- N parameters only, regardless of RKHS dimension. Training data points become the basis centres of the solution automatically.
- A bias term can be added: $\tilde{f}(\boldsymbol{x}) = \sum_n \theta_n \kappa(\boldsymbol{x}, \boldsymbol{x}_n) + b$.
- Regularization is *essential*: without it, infinitely many functions interpolate any finite dataset.

KRR: Problem Formulation

Assume the noisy regression model:

$$y_n = f(\boldsymbol{x}_n) + \varepsilon_n, \quad n = 1, \ldots, N, \quad f \in \mathbb{H}.$$

Adopt squared loss and RKHS-norm regularization:

$$\min_{f \in \mathbb{H}} \sum_{n=1}^{N} \big(y_n - f(\boldsymbol{x}_n)\big)^2 + C \, \|f\|_{\mathbb{H}}^2.$$

By the representer theorem, substitute $f(\boldsymbol{x}) = \sum_{m=1}^{N} \theta_m \kappa(\boldsymbol{x}, \boldsymbol{x}_m)$ and minimise over $\boldsymbol{\theta} \in \mathbb{R}^N$.

KRR: Step-by-Step Derivation

Let $K \in \mathbb{R}^{N \times N}$ with $K_{ij} = \kappa(\boldsymbol{x}_i, \boldsymbol{x}_j)$. Substituting the representer expansion:

$$J(\boldsymbol{\theta}) = (\boldsymbol{y} - K\boldsymbol{\theta})^T(\boldsymbol{y} - K\boldsymbol{\theta}) + C\,\boldsymbol{\theta}^T K\boldsymbol{\theta}.$$

Set $\nabla_{\boldsymbol{\theta}} J = \mathbf{0}$:

$$-2K(\boldsymbol{y} - K\boldsymbol{\theta}) + 2CK\boldsymbol{\theta} = \mathbf{0} \implies (K + CI)\hat{\boldsymbol{\theta}} = \boldsymbol{y}.$$

Closed-form solution:

$$\hat{\boldsymbol{\theta}} = (K + CI)^{-1}\boldsymbol{y}.$$

(Uses $K^T = K$; invertibility guaranteed for most standard kernels.)

KRR: Predictor and Complexity

Define $\boldsymbol{\kappa}(\boldsymbol{x}^*) := [\kappa(\boldsymbol{x}^*, \boldsymbol{x}_1), \ldots, \kappa(\boldsymbol{x}^*, \boldsymbol{x}_N)]^T$.
KRR predictor:

$$\hat{y}(\boldsymbol{x}^*) = \boldsymbol{y}^T (K + CI)^{-1} \boldsymbol{\kappa}(\boldsymbol{x}^*).$$

- Training cost: $O(N^3)$ to compute $(K + CI)^{-1}$.
- Prediction cost: $O(N)$ per test point after training.
- Regularization effect: larger $C \Rightarrow$ smoother f; $C \to 0 \Rightarrow$ exact interpolation; $C \to \infty \Rightarrow$ zero predictor.

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GP Posterior: Setup

Place a GP prior and Gaussian noise on the training model:

$$f \sim \mathcal{GP}(0, \kappa(\cdot,\cdot)), \quad \boldsymbol{y} = \boldsymbol{f} + \varepsilon, \quad \varepsilon \sim \mathcal{N}(\boldsymbol{0}, \sigma_n^2 I).$$

The joint covariance of observations is $K_y := K + \sigma_n^2 I$.
GP posterior at test point $\boldsymbol{x}^*$Lecture :

$$m(\boldsymbol{x}^*) = \boldsymbol{\kappa}^T K_y^{-1} \boldsymbol{y}, \qquad \mathbb{V}(\boldsymbol{x}^*) = \kappa(\boldsymbol{x}^*, \boldsymbol{x}^*) - \boldsymbol{\kappa}^T K_y^{-1} \boldsymbol{\kappa},$$

where $\boldsymbol{\kappa} := [\kappa(\boldsymbol{x}^*, \boldsymbol{x}_1), \ldots, \kappa(\boldsymbol{x}^*, \boldsymbol{x}_N)]^T$.

KRR = GP Posterior Mean

	KRR	**GP posterior**
Prior	RKHS norm $\|f\|_{\mathbb{H}}^2$	GP prior κ
Regularisation	C	σ_n^2 (noise variance)
Predictor	$\boldsymbol{\kappa}^T(K + CI)^{-1}\boldsymbol{y}$	$\boldsymbol{\kappa}^T K_y^{-1}\boldsymbol{y}$
Uncertainty	posterior variance	posterior variance

Key identity: with $C = \sigma_n^2$, KRR predictor = GP posterior mean.
KRR is the MAP solution under a GP prior; GP additionally quantifies uncertainty.

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Take-Home Messages

1. Kernel trick: replace explicit $\phi(\boldsymbol{x})$ by $\kappa(\boldsymbol{x}, \boldsymbol{x}')$ to operate in possibly infinite-dimensional feature spaces cheaply.
2. RKHS: the reproducing property links inner products to function evaluations; positive-definiteness of κ characterises valid kernels.
3. Representer theorem: any regularised RKHS problem has a finite-dimensional solution $f^* = \sum_n \theta_n \kappa(\cdot, \boldsymbol{x}_n)$.
4. KRR: squared-loss + RKHS-norm regularisation gives $(K + CI)\hat{\boldsymbol{\theta}} = \boldsymbol{y}$ and predictor $\hat{y}(\boldsymbol{x}^*) = \boldsymbol{y}^T (K + CI)^{-1} \boldsymbol{\kappa}(\boldsymbol{x}^*)$.
5. KRR = GP MAP: the KRR predictor equals the GP posterior mean when $C = \sigma_n^2$; GP further provides uncertainty estimates.

The End